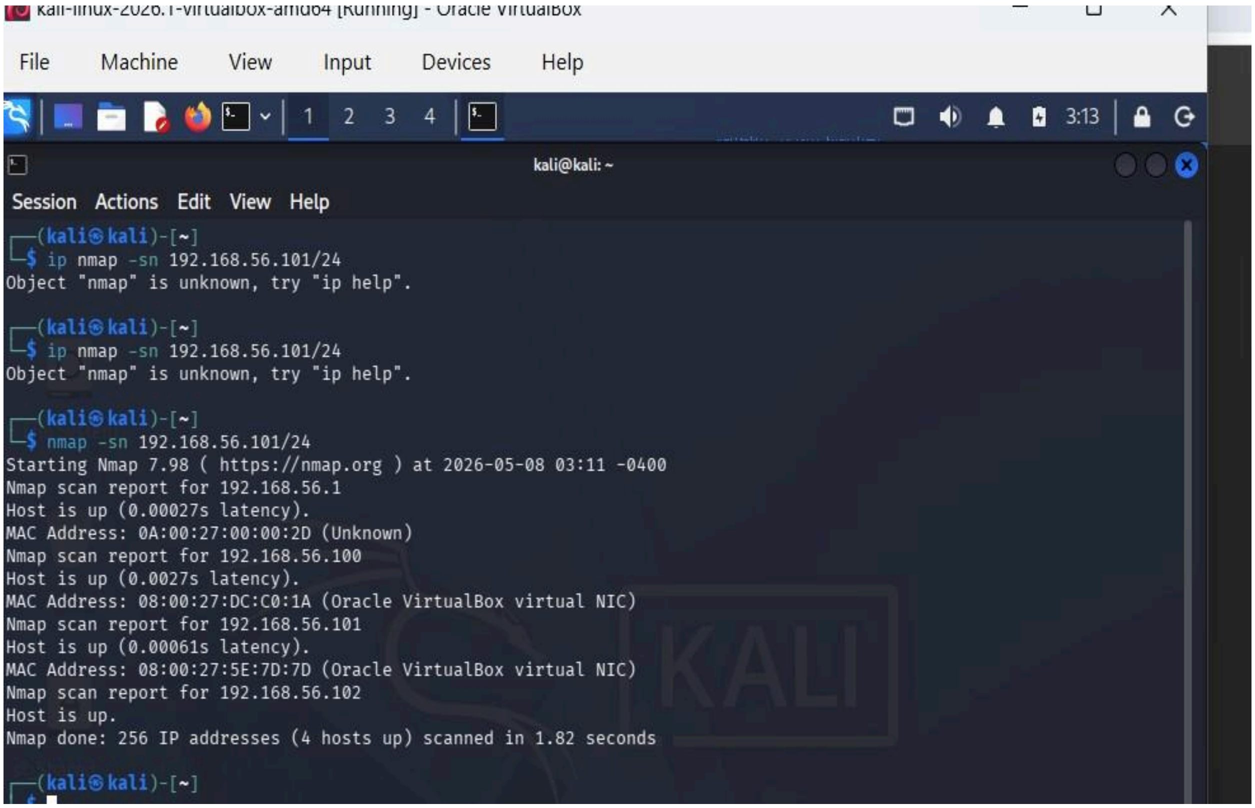
Easy – Network Topology Discovery
Objective: Discover live hosts and listening services.
Deliverable: Screenshot of nmap -sn output + ss -tulpn output.



How many hosts did you discover?
=>4

```
┌──(kali㉿kali)-[~]
└─$ ss -tulpn
Netid      State      Recv-Q      Send-Q          Local Address:Port          Peer Address:Port     Process

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start ssh
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
     Active: active (running) since Fri 2026-05-08 03:17:54 EDT; 22s ago
 Invocation: 85498966dbb14b0daa3a012df28eb2f5
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 6905 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6908 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 2M (peak: 2.8M)
        CPU: 42ms
     CGroup: /system.slice/ssh.service
             └─6908 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

May 08 03:17:54 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 08 03:17:54 kali sshd[6908]: Server listening on 0.0.0.0 port 22.
May 08 03:17:54 kali sshd[6908]: Server listening on :: port 22.
May 08 03:17:54 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

┌──(kali㉿kali)-[~]
└─$ ss -tulpn
Netid      State      Recv-Q      Send-Q          Local Address:Port          Peer Address:Port     Process
tcp        LISTEN     0           128                   0.0.0.0:22                 0.0.0.0:*
tcp        LISTEN     0           128                      [::]:22                    [::]:*

┌──(kali㉿kali)-[~]
```

What services are listening on Kali? ·
=>SSH Service

What is the MAC address of your Windows VM?
=>08-00-27-5E-7D-7D

TCP Three-Way Handshake Analysis

Objective: Capture and identify the SYN, SYN-ACK, ACK handshake

Deliverable: Screenshot of the tcpdump -r output with SYN, SYN-ACK, ACK clearly labeled.

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -w handshake.pcap host 192.168.56.101
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C5 packets captured
5 packets received by filter
0 packets dropped by kernel

┌──(kali㉿kali)-[~]
└─$ tcpdump -r handshake.pcap -n

reading from file handshake.pcap, link-type EN10MB (Ethernet), snapshot length 262144
03:53:33.775034 IP 192.168.56.102.36366 > 192.168.56.101.445: Flags [S], seq 2663119864, win 64240, options [mss 14
60,sackOK,TS val 2722753692 ecr 0,nop,wscale 8], length 0
03:53:33.775999 IP 192.168.56.101.445 > 192.168.56.102.36366: Flags [S.], seq 2103036429, ack 2663119865, win 65535
, options [mss 1460,nop,wscale 8,nop,nop,sackOK], length 0
03:53:33.776039 IP 192.168.56.102.36366 > 192.168.56.101.445: Flags [.], ack 1, win 251, length 0
03:53:38.829744 ARP, Request who-has 192.168.56.101 tell 192.168.56.102, length 28
03:53:38.831459 ARP, Reply 192.168.56.101 is-at 08:00:27:5e:7d:7d, length 46
```

Intermediate – Protocol Analysis

Objective: Compare ICMP, TCP (HTTP), and UDP (DNS).

Deliverable: Documentation (bullet points or table) + screenshots of each command output.

```
┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=128 time=0.890 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=128 time=1.02 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=128 time=1.92 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=128 time=1.60 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.890/1.358/1.924/0.422 ms

┌──(kali㉿kali)-[~]
└─$
```

What port does it use?
=>No port is used
Is it connection-oriented or connectionless?
Connectionless
What application layer protocol rides on top?
ICMP

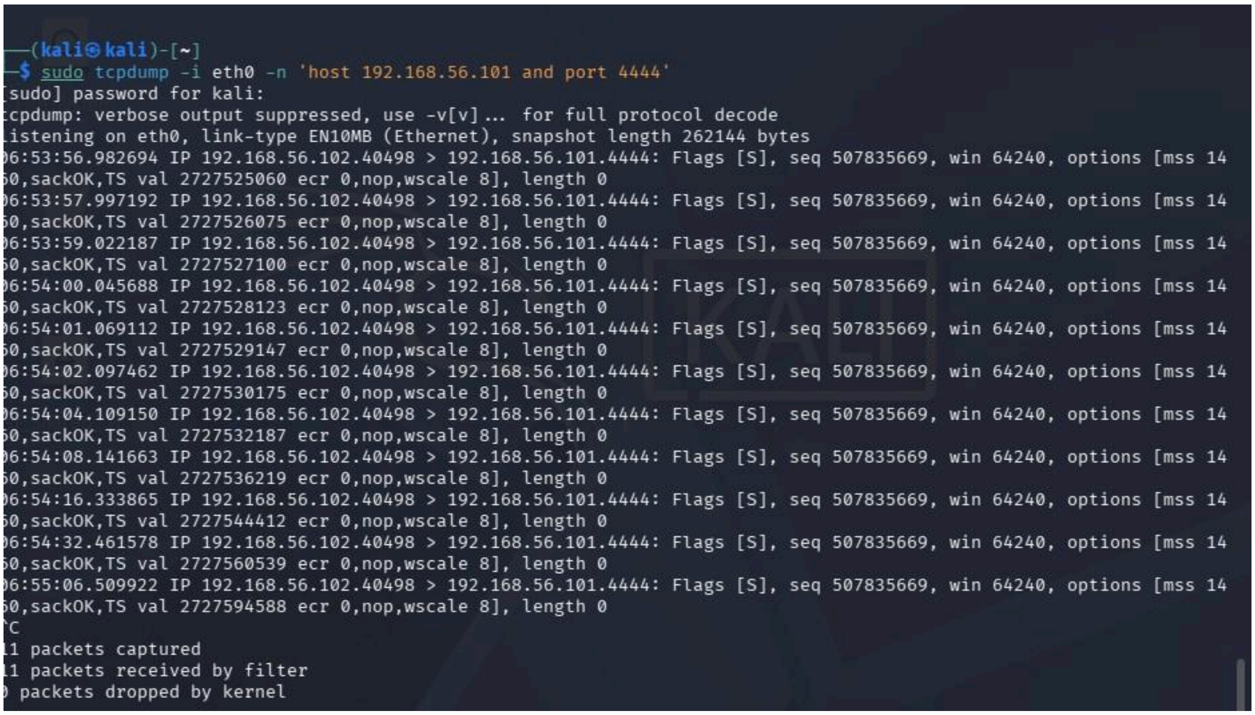
```
┌──(kali㉿kali)-[~]
└─$ curl http://192.168.56.101:8080
^C

┌──(kali㉿kali)-[~]
└─$ curl http://192.168.56.101:80
curl: (28) Failed to connect to 192.168.56.101 port 80 after 133595 ms: Could not connect to server

┌──(kali㉿kali)-[~]
└─$
```

What port does it use?
=>TCP 80
Is it connection-oriented or connectionless?
Connection-oriented
What application layer protocol rides on top?
HTTP

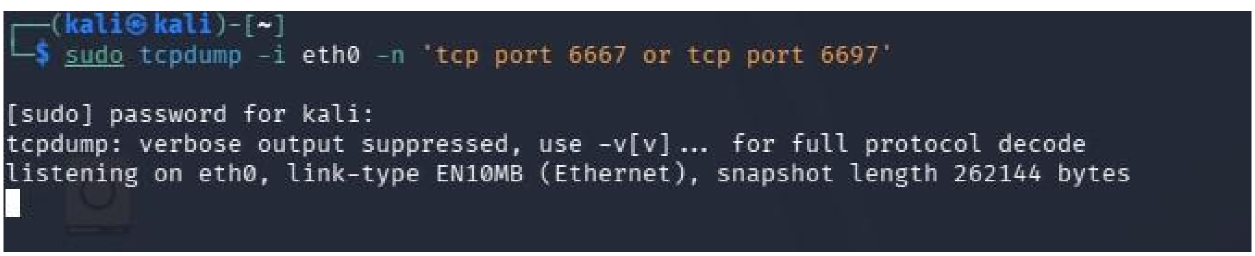
```
┌──(kali㉿kali)-[~]
└─$ nslookup google.com 8.8.8.8
;; UDP setup with 8.8.8.8#53(8.8.8.8) for google.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 8.8.8.8#53(8.8.8.8) for google.com failed: network unreachable.
;; no servers could be reached
;; UDP setup with 8.8.8.8#53(8.8.8.8) for google.com failed: network unreachable.
;; no servers could be reached
```

What port does it use?
=>UDP 53
Is it connection-oriented or connectionless?
Connectionless
What application layer protocol rides on top?
DNS

Expert – Detect Suspicious Connections
Objective: Simulate and detect a malicious reverse shell connection.
Deliverable: Screenshot showing the suspicious connection (port 4444) captured in tcpdump

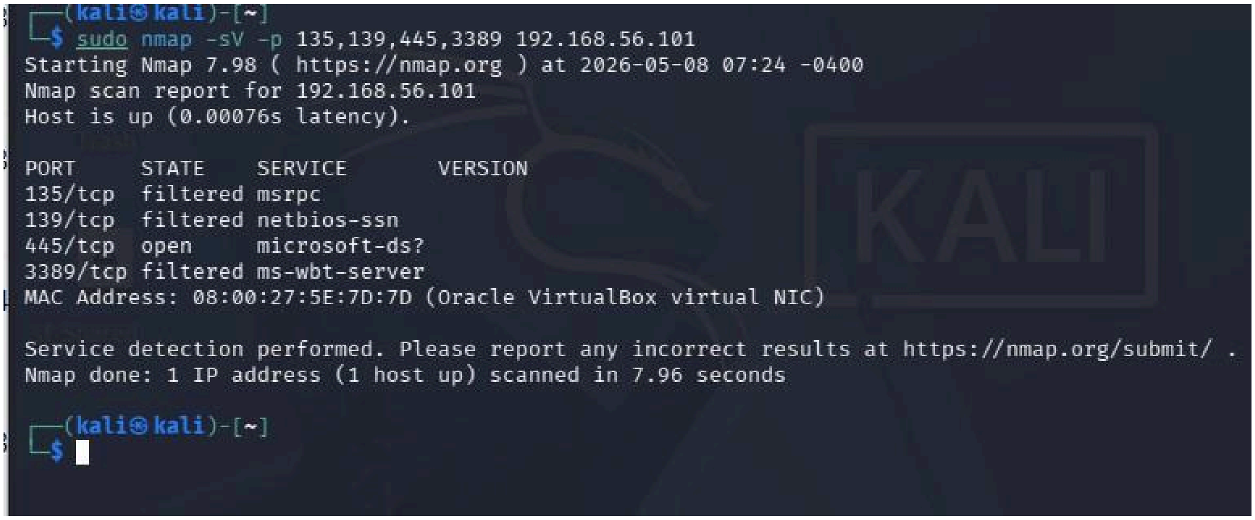


```
──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -n 'host 192.168.56.101 and port 4444'
[sudo] password for kali:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
06:53:56.982694 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727525060 ecr 0,nop,wscale 8], length 0
06:53:57.997192 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727526075 ecr 0,nop,wscale 8], length 0
06:53:59.022187 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727527100 ecr 0,nop,wscale 8], length 0
06:54:00.045688 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727528123 ecr 0,nop,wscale 8], length 0
06:54:01.069112 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727529147 ecr 0,nop,wscale 8], length 0
06:54:02.097462 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727530175 ecr 0,nop,wscale 8], length 0
06:54:04.109150 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727532187 ecr 0,nop,wscale 8], length 0
06:54:08.141663 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727536219 ecr 0,nop,wscale 8], length 0
06:54:16.333865 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727544412 ecr 0,nop,wscale 8], length 0
06:54:32.461578 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727560539 ecr 0,nop,wscale 8], length 0
06:55:06.509922 IP 192.168.56.102.40498 > 192.168.56.101.4444: Flags [S], seq 507835669, win 64240, options [mss 14
60,sackOK,TS val 2727594588 ecr 0,nop,wscale 8], length 0
^C
11 packets captured
11 packets received by filter
0 packets dropped by kernel
```



```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -i eth0 -n 'tcp port 6667 or tcp port 6697'

[sudo] password for kali:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
```

Expert – Service Fingerprinting
Objective: Identify services and their versions on the Windows VM.
Deliverable: Completed table + screenshot of nmap -sV output.



Intermediate – Network Mapping Script.

Objective: Automate network discovery with a Bash script.

Deliverable: · The script (copy/paste in submission) ·

Screenshot of execution showing discovered hosts.

```
#!/bin/bash
# network_mapper.sh - GraySentinel Network Discovery
NETWORK=${1:-"192.168.56.0/24"}
OUTPUT_DIR="GraySentinel_Commissioning/1_Portfolio/Reports/"
mkdir -p "$OUTPUT_DIR"
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
OUTPUT_FILE="${OUTPUT_DIR}network_map_${TIMESTAMP}.txt"
echo "[+] Scanning network: $NETWORK"
echo "Network Discovery Report - $(date)" > "$OUTPUT_FILE"
echo "================================" >> "$OUTPUT_FILE"
# Ping sweep
echo -e "\n[+] Live Hosts:" >> "$OUTPUT_FILE"
nmap -sn "$NETWORK" | grep "Nmap scan report" | awk '{print $5}' >> "$OUTPUT_FILE"
# Service scan
echo -e "\n[+] Service Detection:" >> "$OUTPUT_FILE"
nmap -sV --open "$NETWORK" >> "$OUTPUT_FILE"
echo "[+] Report saved to $OUTPUT_FILE"
```

```
Session  Actions  Edit  View  Help
 GNU nano 8.7.1                                      network_mapper.sh
#!/bin/bash
# network_mapper.sh - GraySentinel Network Discovery
NETWORK=${1:-"192.168.56.0/24"}
OUTPUT_DIR="GraySentinel_Commissioning/1_Portfolio/Reports/"
mkdir -p "$OUTPUT_DIR"
TIMESTAMP=$(date +"%Y%m%d_%H%M%S")
OUTPUT_FILE="${OUTPUT_DIR}network_map_${TIMESTAMP}.txt"
echo "[+] Scanning network: $NETWORK"
echo "Network Discovery Report - $(date)" > "$OUTPUT_FILE"
echo "================================" >> "$OUTPUT_FILE"
# Ping sweep
echo -e "\n[+] Live Hosts:" >> "$OUTPUT_FILE"
nmap -sn "$NETWORK" | grep "Nmap scan Report" | awk '{print $5}' >> "$OUTPUT_FILE"
# Service scan on found hosts
echo -e "\n[+] Service Detection:" >> "$OUTPUT_FILE"
nmap -sV --open "$NETWORK" >> "$OUTPUT_FILE"
echo "[+] Report saved to $OUTPUT_FILE"
```

```
└─$ ./network_mapper.sh

[+] Scanning network: 192.168.56.0/24
[+] Report saved to GraySentinel_Commissioning/1_Portfolio/Reports/network_map_20260508_075354.txt

┌──(kali㉿kali)-[~]
└─$ cat GraySentinel_Commissioning/1_Portfolio/Reports/network_map_*.txt
Network Discovery Report - Fri May  8 07:44:07 AM EDT 2026
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[+] Live Hosts:
Network Discovery Report - Fri May  8 07:50:13 AM EDT 2026
═════════════════════════════════

[+] Live Hosts:
Network Discovery Report - Fri May  8 07:52:00 AM EDT 2026
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[+] Live Hosts:

[+] Service Detection:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-08 07:52 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0014s latency).
Not shown: 999 filtered tcp ports (no-response)
Some closed ports may be reported as filtered due to --defeat-rst-ratelimit
PORT    STATE SERVICE       VERSION
445/tcp open  microsoft-ds?
MAC Address: 08:00:27:5E:7D:7D (Oracle VirtualBox virtual NIC)

Network Discovery Report - Fri May  8 07:53:54 AM EDT 2026
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[+] Live Hosts:

[+] Service Detection:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-08 07:53 -0400
Nmap scan report for 192.168.56.101
Host is up (0.0015s latency).
Not shown: 999 filtered tcp ports (no-response)
```